

ISI

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Gross planned investment (I) = 7. Share of consumption in the aggregate planned demand (E) is $\frac{1}{2}$ when $Y = 0$. Further, when $Y = 30$, unplanned inventory accumulation is 5. Compute the autonomous expenditure multiplier in this model.

8. A firm is contemplating an investment of Rs.200 in a project. The project yields a revenue flow of Rs. 100 per year from next year onwards. However, each year there is some probability p (given exogenously) that the project might be abandoned forever. Assuming $p = \frac{1}{4}$ and a rate of discount of 10% per year, compute whether it is profitable for the firm to undertake the project.
9. A firm has a production function $y = x_1 x_2$, where y is the amount of its output and x_1 and x_2 are the amounts of the two inputs used. Suppose, prices of these two inputs are given at r_1 and r_2 . If the minimum cost of production of this firm is 4, what is the value of y ?

R3

Test code: ME I/ ME II, 2003

Syllabus for ME I

Subscript
2003

Matrix Algebra : Matrices and Vectors, Matrix Operations, Determinants, Nonsingularity, Inversion, Cramer's rule.

Calculus : Limits, Continuity, Differentiation of functions of one or more variables, Product rule, Partial and total derivatives, Derivatives of implicit functions, Unconstrained optimization (first and second order conditions for extrema of a single variable and several variables). Taylor Series, Definite and Indefinite Integrals: standard formulae, integration by parts and integration by substitution, Differential equations, Constrained optimization of functions of a single variable.

Theory of Sequence and Series:

Linear Programming: Formulations, statements of Primal and Dual problems, Graphical Solutions.

Theory of Polynomial Equations (up to third degree).

Elementary Statistics: Measures of central tendency, dispersion, correlation, Elementary probability theory.

Sample Questions for ME I (Mathematics), 2003

1. The function $f(x) = |x|$ is

- (A) both continuous and differentiable at $x = 0$;
 (B) neither continuous nor differentiable at $x = 0$;
 (C) continuous, but not differentiable at $x = 0$;
 (D) differentiable, but not continuous at $x = 0$.

2. Let $f(x)$ be a function satisfying the condition $f(x) = f(-x)$ for all real x . If $f'(0)$ exists, then $f'(0)$ will be equal to

- (A) 0; (B) -1; (C) 1; (D) none of these.

3. $\int_0^1 \frac{x+x^2+x^3+\dots+x^n-n}{x-1} dx$ is equal to

- (A) n^2 ; (B) $\frac{n+1}{2}$; (C) $\frac{n(n+1)}{2}$; (D) none of the above.

4. Suppose a time-dependent variable, $X(t)$, changes over time according to $\frac{dX(t)}{dt} = 3t^{1/2}$. Then, at any time t , $X(t)$ depends on its initial value $X(0)$ in the following way:

- (A) $X(0) + 3t^{1/2}$; (B) $X(0) + 3t^{3/2}$;
 (C) $X(0) + 2t^{3/2}$; (D) $X(0) + 2t^{1/2}$.

5. The second-order Taylor expansion of $\frac{1}{1+x}$ around $x_0 = 1$ is:

- (A) $\frac{1}{2} + \frac{1}{2(1+x)} + \text{remainder}$; (B) $\frac{1}{2} - \frac{1}{8}(x-1) + \frac{1}{4}(x-1)^2 + \text{remainder}$;
 (C) $\frac{1}{2} + \frac{1}{2}(x-1) + \frac{1}{8}(x-1)^2 + \text{remainder}$;
 (D) $\frac{1}{2} - \frac{1}{4}(x-1) + \frac{1}{8}(x-1)^2 + \text{remainder}$.

6. If A is set of all real numbers such that $|1-x| - x \geq 0$, then A equals

- (A) $[-\frac{1}{2}, \frac{1}{2}]$; (B) $(-\infty, 0]$;
 (C) $(-\infty, -\frac{1}{2}]$; (D) none of the above.

7. For an AP series it is given that $\frac{S_m}{S_n} = \frac{m^2}{n^2}$, where S_m (S_n) is the sum upto

m (n) terms. Then $\frac{t_m}{t_n}$, where t_m stands for the m -th term, equals

- (A) $\frac{2m-1}{2n-1}$; (B) $\frac{2m+1}{2n+1}$; (C) $\frac{2m-1}{2n+1}$; (D) $\frac{2m+1}{2n-1}$.

8. The expression $\sqrt{11+6\sqrt{2}} + \sqrt{11-6\sqrt{2}}$ is

- (A) an even positive integer;
 (B) an even positive number with an even decimal component;
 (C) an odd positive number with an odd decimal component;
 (D) none of the above.

9. The problem, $\max (2x^2 + y^2)$ subject to $x + y \leq 1$; $x, y \geq 0$, has a solution in which the value of x is given by

- (A) $x = \frac{1}{2}$; (B) $x = \frac{2}{3}$;
 (C) $x = 1$; (D) none of the above.

10. In what ratio should a given line be divided into two parts so that the area of the rectangle contained by them is maximum?

- (A) 1 : 1; (B) 4 : 3;
 (C) 3 : 2; (D) $(\sqrt{5} + 1) : (\sqrt{5} - 1)$.

11. The derivative of $\int_a^{100} x dx$ with respect to a is:

- (A) a ;
 (B) $-a$;
 (C) a^2 ;
 (D) $100a$.

12. The domain of the function $f(x) = \sqrt{x^2 - |x|} + \frac{1}{\sqrt{9 - x^2}}$ is

- (A) $(-2, -1)$; (B) $[2, 3)$; (C) $\{0\}$; (D) none of the above .

13. The range of the function $f(x) = \sqrt{-4x^2 + 4x + 3}$ is

- (A) $[0, -1)$; (B) $[1, 3]$; (C) $(0, 1)$;
 (D) none of the above .

14. What is the sum of the series $1^2 + 2^2 x + 3^2 x^2 + 4^2 x^3 + \dots$, where $0 < x < 1$?

- (A) 1 ;
 (B) $(1-x)^{-1}$;
 (C) $(1+2x)(1+x)^{-1}$;
 (D) $(1+x)(1-x)^{-2}$.

15. $F(y) = \int_0^1 x^2 y dx$ and $G(y) = y + |y|$. Then

- (A) $F(y) > G(y)$ for all $y \geq 0$;
 (B) $F(y) < G(y)$ for all $y \geq 0$;
 (C) $F(y) < G(y)$ for all $y > 0$;

(D) none of the above .

16. The function $\sqrt{4x - x^2}$ is

- (A) bounded whenever the function is defined ;
 (B) unbounded ;
 (C) well-defined for all values of x ;
 (D) none of the above .

17. Consider the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $ad \neq bc$. Then the

expression, $\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, is :

- (A) the right inverse of A ; (B) the left inverse of A ;
 (C) the inverse of A ; (D) none of the above .

18. Let X and Y be two random variables having $\text{Var}(X) = 4$ and $\text{Var}(Y) = 9$. Let V be the variance of $(X - Y)$, when X and Y are Independent and R be the variance of $(X - Y)$, when $\text{Cov}(X, Y) = 2$. Then we have

- (A) $V > R$; (B) $V < R$; (C) $V = R$;
 (D) Cannot be said definitely .

Syllabus for ME II (Economics)

Microeconomics: Theory of consumer behaviour, Theory of producer behaviour, Market forms and Welfare economics.

Macroeconomics: National income accounting, Simple model of income

determination and Multiplier, IS-LM model, Aggregate demand and aggregate supply model, Money, Banking and Inflation.

Sample questions for ME II (Economics), 2003

1. Instruction for question numbers 1 (i) - 1 (iv)

For each of the following questions four alternative answers are provided. Choose the answer that you consider to be the most appropriate for a question and write it in your answer book.

- (i) Veena always consumes 3 times as much y as she consumes x . She will consume these goods in no other ratio. The price of x is 3 times the price of y . Veena has an endowment of 18 units of x and 78 units of y which she can trade at the going prices. She has no other source of income. How much x will Veena buy from the market?
 (A) 8; (B) 6; (C) 4; (D) the answer cannot be determined without knowing the price of x .
- (ii) Let an agent's utility function be $U\left(C, \frac{M}{P}\right)$ which depends on consumption (C) and real money balances $\left(\frac{M}{P}\right)$, where P is the price of C and M is the nominal balances demanded. The agent's budget constraint is $PC + M = \bar{M}$, where $\bar{M}$ is the given amount of money balance in her possession. The substitution effect of a rise in P on C for this agent is ;
 (A) zero; (B) -1 ;
 (C) indeterminate; (D) none of the above.

- (iii) Consider a monopolist with no fixed cost of production. Its marginal cost of production is 5. The demand for the product is

$$D(p) = \begin{cases} \frac{10}{p} & \text{if } p \leq 10 \\ 0 & \text{if } p > 10 \end{cases}$$

Assume that no fractional unit of output can be produced. Then the equilibrium output is:

- (A) 0; (B) 1; (C) 5; (D) 10.
- (iv) This is a problem on national income accounting. Suppose in an economy in a given year the production of capital goods increases by Rs. 100, of which Rs. 80 worth of goods can be sold in the market. In the same year there takes place a simultaneous decline in the import of Rs. 30 worth of capital goods. Everything else remains unchanged. Following these GDP increases by Rs.
 (A) 80; (B) 130; (C) 70; (D) 100.
2. A consumer consumes two commodities X and Y . Price of Y is given at P_y . The commodity X is available at two given prices, P_x and P_x' with $P_x' < P_x$. However X is available at P_x' only in a few shops and hence the buyer has to devote some time (say, t hours a day) to search for these shops so as to get X at the lower price. Suppose, the consumer works for 8 hours a day at a (given) wage rate of w per hour.
- (i) Draw the budget set of the consumer. Show the upper contour of the budget set and the equilibrium of the consumer buying at the lower price.
- (ii) Now suppose that w goes up. Is it possible for the

consumer to decide to buy X at the higher price?

- ③ A cell phone company A charges a fixed fee of Rs. 500 and, in addition to that, Rs. 2 per call. There is another cell phone company B which charges a fixed fee of Rs. 1000 plus Re. 1 per call. There are three consumers, each earning the same income. The first consumer opts for the offer given by the company A, the second opts for the offer given by the company B, while the third chooses none.

- Draw the budget contour and appropriate indifference curves to describe the choice of the three consumers.
- Is it possible for the third consumer to have a utility function of the Cobb-Douglas type?

4. A buyer possessing a 50 - rupee - note goes to a market where only two goods, rice and pulses, are sold. The rice seller offers to give change, if a minimum amount of 3 kg. is bought while the person selling pulses refuses to give any change.

- Draw the budget line of the buyer.
- Indicate the equilibrium purchase of the buyer (in the diagram) assuming that he consumes rice and pulses in a fixed proportion.
- Suppose there is a person sitting outside the market offering to give change for Rs. 50. What is the maximum amount he can charge for this service?
- Suppose the price of pulses increases. Find the effect of this on the equilibrium purchase of the buyer and on the price charged by the money changer?

- ⑤ Consider a society where people throw garbage on the streets. A person who does so inflicts negative externalities on others. Suppose the rupee cost of this externality to any one is an , where a is a positive constant and n is the number of (other) people who throw garbage on the streets. Assume that your own act of throwing garbage on the streets gives you a gain worth G rupees, but requires you to incur a cost of $\left(\frac{C}{n}\right)$ rupees, where C is a positive constant. Show that

- if no one throws garbage on the streets, it is optimal for you not to throw garbage on the streets;
- if everyone else throws garbage on the streets, then there is a population size n^* , such that if n , the number of other people throwing garbage on the streets, is greater than or equal to n^* , it is optimal for you to throw garbage on the streets. What does n^* depend on?

6. Consider the IS function, wherein consumption depends on income only, investment depends on interest rate only and government spending is exogenous. Suppose that, as a consequence of a unit increase in government expenditure, the IS curve shifts vertically by 0.5 at each value of GDP. What is the rate of change of investment with respect to the rate of interest?
7. Consider a Simple Keynesian model for a closed economy without any government. The following data are given for this economy (where Y stands for GDP and aggregate planned demand, E , is a function of Y);